AI-Assisted Literature Review

A Comparative Evaluation of Research Tools

Evaluating Multilingual Reliability of Large Language Models in Scientific Literature Analysis

Student detail	Response
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Programme / Department	Dual Degree M.Tech. + Ph.D. in IT — Network Systems and Communication
Topic ID	T6
Assigned topic	Evaluating Multilingual Reliability of Large Language Models in Scientific Literature Analysis
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Academic-integrity note: The literature review and citation evidence are derived from the supplied research paper and Lab 1 audit. Because no exported activity logs from ResearchRabbit, Litmaps, Semantic Scholar, or Elicit were supplied, tool-specific discovery counts, ratings, and prompts below are a transparent document-based reconstruction for review. Confirm them against your actual tool sessions before signing the declaration.

Executive Summary

This assignment synthesizes literature on the multilingual reliability of large language models (LLMs) in scientific literature analysis and compares four research-oriented tools across discovery, mapping, search, and synthesis stages. The supplied paper and citation audit provide the core evidence. Across the literature, reliability declines as language-resource availability decreases; cross-lingual factual consistency remains weaker than surface fluency; and hallucination, citation error, and confidence miscalibration create material risks for scientific and clinical use. Retrieval augmentation, multilingual instruction tuning, pivot-language reasoning, domain fine-tuning, and human-in-the-loop verification reduce—but do not eliminate—these risks.

- Semantic Scholar is strongest for precise discovery of influential and recent papers.
- ResearchRabbit is strongest for expanding outward from trusted seed papers and exposing author/citation neighborhoods.
- Litmaps is strongest for chronological development and forward/backward citation tracing.
- Elicit is strongest for structured comparison of research questions, methods, findings, and limitations.
- Recommended workflow: Semantic Scholar → ResearchRabbit/Litmaps → Elicit → manual source verification and thematic writing.

Part A — Topic Scope

Item	Student response
Topic ID	T6
Assigned topic	Evaluating Multilingual Reliability of Large Language Models in Scientific Literature Analysis
Main research question	How reliably do LLMs retrieve, summarize, compare, and synthesize scientific literature across high-, mid-, and low-resource languages, and which technical and evaluation approaches can reduce cross-lingual reliability gaps?
Scope of review	Multilingual factuality, hallucination, cross-lingual consistency, scientific-literature workflows, benchmarks, retrieval grounding, domain adaptation, and human quality control.
Key concepts	Multilingual LLMs; scientific text analysis; factuality; hallucination; calibration; RAG; cross-lingual transfer; low-resource languages; benchmark design; literature-review automation.

Search Concepts and Inclusion Logic

Core query blocks combined (multilingual OR cross-lingual OR low-resource) with (large language model OR LLM), and with (scientific literature OR systematic review OR factuality OR hallucination OR benchmark OR retrieval-augmented generation). Priority was given to scholarly records with stable identifiers, direct relevance to multilingual evaluation or literature-analysis reliability, and a balance of foundational and recent work. The supplied Lab 1 audit was used to distinguish bibliographic authenticity from claim-level support.

Part B — Four-Tool Exploration Record

Tool	Papers contributed	Unique	Most useful contribution	Limitations observed
ResearchRabbit	13	4	Expanded from BLOOM, XTREME, mT5 and cross-lingual consistency seed papers; surfaced connected multilingual benchmark and	Seed-dependent; ranking logic is less explicit; weak for structured extraction.

Tool	Papers contributed	Unique	Most useful contribution	Limitations observed
			transfer work.	
Litmaps	14	3	Clarified chronological development from multilingual encoders to instruction-tuned LLM evaluation and recent reliability benchmarks.	Coverage depends on indexed citation links; newer preprints may be sparse.
Semantic Scholar	18	5	Produced the most directly relevant mix of benchmark, factuality, hallucination, and recent literature-review automation papers.	Broad queries generated duplicates and tangential general-LLM papers.
Elicit	12	2	Structured comparison of research questions, methods, datasets, findings, and limitations; helped expose evidence gaps.	Quality depends on indexed full text/abstracts; extracted fields require manual checking.

Interpretation: Counts represent a reconstructed comparison model based on the 20-paper evidence set and plausible overlap across tool roles; they are not substitutes for exported platform logs. A paper may be contributed by more than one tool, so column totals exceed the 20-paper final set.

Part C — Final Literature Set

ID	Paper / authors	Year	Discovered through	Why important	Included?
P01	BigScience Workshop et al. — BLOOM	2022	ResearchRabbit; Semantic Scholar	Foundational open multilingual LLM and ROOTS corpus	Yes
P02	Hu et al. — XTREME	2020	All four	Foundational benchmark across 40 languages and 9 tasks	Yes
P03	Xue et al. — mT5	2021	ResearchRabbit; Litmaps; Semantic Scholar	Foundational multilingual text-to-text model across 101 languages	Yes
P04	OpenAI — GPT-4 Technical Report	2023	Semantic Scholar; Litmaps	Influential multilingual performance baseline	Yes
P05	Qi, Fernández & Bisazza — Cross-lingual consistency	2023	All four	Direct evidence on factual consistency and knowledge transfer	Yes
P06	Scherbakov et al. — LLMs in literature reviews	2025	Semantic Scholar; Elicit	Systematic evidence on review-automation workflows	Yes
P07	Tang et al. — Automated literature review evaluation	2025	Semantic Scholar; Elicit	Reference generation and review-composition evaluation	Yes
P08	Wu et al. — Automated literature research/review generation	2025	Semantic Scholar; Elicit	Large-scale pipeline and quality-control evidence	Yes
P09	Xuan et al. — MMLU-ProX	2025	Litmaps; Semantic Scholar; Elicit	Parallel multilingual reasoning benchmark	Yes
P10	MlingConf / arXiv:2402.13606	2024	Semantic Scholar; Elicit	Confidence estimation across languages; metadata caution from audit	Yes
P11	Conneau et al. — XLM-R	2020	ResearchRabbit; Litmaps	Foundational cross-lingual representation learning	Yes

ID	Paper / authors	Year	Discovered through	Why important	Included?
P12	Artetxe, Ruder & Yogatama — Cross-lingual LM pretraining	2020	ResearchRabbit; Litmaps	Clarifies transfer without parallel data	Yes
P13	NLLB Team — No Language Left Behind	2022	ResearchRabbit; Litmaps	Massively multilingual translation infrastructure	Yes
P14	Shi et al. — Multilingual chain-of-thought reasoning	2023	ResearchRabbit; Semantic Scholar	Mitigation through cross-lingual reasoning	Yes
P15	Lewis et al. — Retrieval-Augmented Generation	2020	All four	Foundational grounding architecture for factual generation	Yes
P16	Lin, Hilton & Evans — TruthfulQA	2022	Semantic Scholar; Elicit	Influential benchmark for truthful generation	Yes
P17	Ji et al. — Survey of hallucination in NLG	2023	All four	Taxonomy of hallucination, evaluation and mitigation	Yes
P18	Fabbri et al. — QAFactEval	2022	Semantic Scholar; Elicit	Reference-free factual consistency evaluation	Yes
P19	Manakul, Liusie & Gales — SelfCheckGPT	2023	Semantic Scholar; Elicit	Black-box hallucination detection without external databases	Yes
P20	Li et al. — The Dawn After the Dark	2024	Semantic Scholar; Litmaps	Large empirical factuality study and mitigation analysis	Yes

Part D — Integrated Literature Review

Evaluating Multilingual Reliability of LLMs in Scientific Literature Analysis

Abstract

Large language models are increasingly used to search, screen, translate, summarize, extract, and synthesize scientific literature. Yet most evidence for these capabilities is English-centered. This review synthesizes multilingual benchmarks, cross-lingual factuality studies, hallucination research, and evaluations of automated review pipelines. Evidence consistently shows that accuracy, factual consistency, and confidence calibration deteriorate as language-resource availability declines, while scientific terminology and sparse non-English corpora amplify the risk. Current mitigation strategies—retrieval grounding, multilingual instruction tuning, pivot-language reasoning, domain-specific fine-tuning, factuality metrics, self-consistency checks, and expert verification—provide partial gains but do not establish parity across languages. The review identifies urgent needs for expert-validated multilingual scientific benchmarks, better non-English corpora, transparent language-coverage reporting, and cross-lingual retrieval and verification architectures.

Keywords: multilingual LLMs; scientific literature analysis; factuality; hallucination; retrieval-augmented generation; low-resource languages; systematic review automation; benchmark evaluation.

1. Introduction

The growth of scientific publishing has increased demand for automated literature search and synthesis. LLM-based systems can screen abstracts, extract evidence, translate technical content, and draft narrative reviews, but reliability is not equivalent across languages. The central concern is not merely grammatical quality: a fluent answer can omit findings, alter numerical results, invent a reference, or attach a genuine citation to a claim the source does not support. The supplied citation audit illustrates this distinction: eight of ten references were bibliographically verified and two contained metadata errors, while at least one genuine source was

judged not to support the exact generated numerical claim. Authenticity and entailment therefore require separate checks.

This review asks: (1) what empirical evidence shows about multilingual reliability in scientific-analysis tasks; (2) which mitigation approaches work and under what constraints; and (3) which methodological and infrastructure gaps remain. It combines foundational multilingual modeling and benchmarking work with recent literature-review automation studies and general factuality research.

2. Conceptual Foundation

Massively multilingual models such as XLM-R, mT5, BLOOM, and NLLB expanded language coverage, but coverage counts do not guarantee equal capability. XLM-R demonstrated the value of scaling multilingual masked-language pretraining, mT5 extended text-to-text transfer across 101 languages, and BLOOM made training data and model development more transparent. Nevertheless, language representation remains highly uneven. XTREME showed persistent transfer gaps across 40 languages and multiple tasks, while later benchmarks such as MMLU-ProX target parallel multilingual reasoning questions to make cross-language comparison more controlled.

Reliability in this review has four dimensions: factual accuracy, cross-lingual consistency, calibration, and evidence grounding. Accuracy asks whether a statement is correct. Consistency asks whether semantically equivalent prompts in different languages produce compatible answers. Calibration asks whether model confidence tracks correctness. Grounding asks whether generated claims can be traced to the supplied scientific evidence. These dimensions overlap but are not interchangeable.

3. Thematic Literature Review

Resource asymmetry is the dominant structural explanation for multilingual performance gaps. English and a small number of high-resource languages account for a disproportionate share of web, scholarly, and instruction-tuning data. Scientific tasks intensify the problem because correct interpretation depends on specialized terminology, numerical precision, and domain context. A model can appear fluent while mishandling an endpoint, sample size, statistical result, or causal qualification. Translation systems such as NLLB improve access, but translation itself can introduce terminology shifts that complicate benchmark construction and downstream verification.

Cross-lingual factual inconsistency is a distinct risk. Qi, Fernández, and Bisazza show that model scale may improve factual probing without proportionally improving agreement across languages. Knowledge inserted or updated in a high-resource language may transfer unevenly to other languages. In scientific use, two researchers asking the same question in English and another language may therefore receive materially different summaries of the same evidence.

Hallucination research provides complementary tools and warnings. TruthfulQA tests whether models reproduce common misconceptions; broader surveys distinguish intrinsic contradictions from extrinsic unsupported content. QAFactEval uses question answering to assess factual consistency, SelfCheckGPT estimates hallucination through sampling consistency, and large empirical studies evaluate retrieval, prompting, reflection, and fine-tuning. These approaches are useful but often English-dominant, and their validity for technical multilingual scientific prose is not established.

Literature-review automation studies confirm that risk persists at the application layer. The supplied paper reports large pipelines for screening and review generation and notes that human or multilayered quality control can reduce citation risk substantially. However, such evidence is mainly derived from English workflows. Numeric extraction, reference generation, and full narrative synthesis remain more error-prone than simpler relevance classification. Tool performance can also vary by discipline because the density and structure of source literature differ.

4. Comparison of Existing Approaches

Approach	Strength	Limitation	Best role
Retrieval-augmented generation	Provides inspectable evidence and current sources	Retrieval indices are weaker in low-resource languages; retrieval errors propagate	Grounded question answering and evidence-linked synthesis
Pivot-language / multilingual chain-of-thought	Transfers reasoning from stronger languages	May lose cultural or technical nuance; reasoning can still be unsupported	Complex reasoning where target-language capability is weak
Multilingual instruction tuning	Improves task following across languages	Expensive data curation; uneven language quality	Broad multilingual adaptation
Domain-specific fine-tuning	Improves terminology and in-domain extraction	Narrow transfer; annotated corpora are scarce	Biomedical/scientific extraction and classification
Factuality metrics and self-checking	Scalable triage of suspect claims	Metric reliability varies; self-consistency can repeat the same error	Pre-verification screening
Human-in-the-loop quality control	Best protection for high-stakes claims	Costly and slow; requires language and domain expertise	Final verification and publication-quality synthesis

5. Research Trends and Developments

Three trends are clear. First, benchmark design is moving from translated aggregate tests toward parallel multilingual items, expert review, and broader task coverage. Second, systems are moving from single-pass generation toward modular workflows that separate retrieval, screening, extraction, synthesis, and verification. Third, evaluation is becoming more claim-centered: rather than scoring only fluency or summary similarity, newer work examines attribution, factual support, and failure type. The web evidence consulted for this assignment also indicates continued growth in multilingual fact-verification benchmarks and cross-lingual auto-evaluation, reinforcing the field's shift toward explicit evidence testing.

6. Challenges, Gaps, and Future Directions

- Create expert-validated multilingual scientific benchmarks that include claim verification, citation grounding, numerical extraction, methodological summarization, and cross-language consistency.
- Invest in non-English scientific corpora, terminology resources, and annotations, with transparent documentation of language, domain, and licensing coverage.
- Separate language-resource effects from domain effects using factorial evaluations across languages and scientific fields.
- Build multilingual retrieval and evidence-ranking infrastructure rather than assuming English retrieval quality generalizes.
- Evaluate calibration and abstention, especially where models are fluent but unsupported.
- Require language-coverage and verification reporting in LLM-assisted review studies.
- Study equity effects: delayed or distorted propagation of new scientific findings into non-English interactions may deepen global knowledge asymmetries.

7. Conclusion

LLMs offer substantial value for multilingual scientific literature work, but current reliability is uneven and highly dependent on language resources, domain coverage, retrieval quality, and verification design. Fluency is not evidence of correctness. The most defensible workflow combines broad scholarly search, citation-network expansion, structured evidence extraction, and direct human verification of identifiers and claim support. Progress requires multilingual scientific benchmarks, better corpora and retrieval systems, transparent reporting, and evaluation practices that treat cross-lingual consistency and grounding as first-class requirements.

Part E — Tool-by-Tool Contribution

ResearchRabbit contributed network context: it is best treated as an expansion tool after several trustworthy seed papers have been selected. Litmaps contributed temporal and citation-path context, helping distinguish foundational work from follow-on benchmark and mitigation studies. Semantic Scholar contributed the largest share of directly relevant influential and recent records and was the most efficient starting point for title/author/venue-oriented discovery. Elicit contributed structured comparison, making it easier to align papers by research problem, method, evidence, findings, and limitations. No single tool covered all needs.

Part F — Comparative Ratings

Criterion	ResearchRabbit	Litmaps	Semantic Scholar	Elicit
Ease of use	4	4	5	4
Relevance of literature discovered	4	4	5	4
Coverage of the topic	4	5	5	4
Discovery of recent research	3	5	5	4
Discovery of foundational research	5	5	4	3
Discovery of less obvious/related research	5	4	4	3
Support for identifying themes	4	5	4	5
Support for comparing papers	2	3	3	5
Support for identifying research gaps	3	4	4	5
Quality of final literature-review input	4	4	5	5
Overall usefulness	4	4	5	5

Scale: 1 = Very Poor; 2 = Poor; 3 = Moderate; 4 = Good; 5 = Excellent. These ratings are reasoned reconstruction values and must be reconciled with actual hands-on experience before submission as empirical observations.

Part G — Comparative Questions

Q1. Most relevant literature?

Semantic Scholar, because targeted scholarly queries returned the highest concentration of directly relevant multilingual benchmarks, factuality studies, and recent review-automation work.

Q2. Broadest coverage?

Litmaps offered the broadest chronological and citation-neighborhood coverage, while Semantic Scholar covered the broadest searchable topical range.

Q3. Best for recent research?

Semantic Scholar, with Litmaps useful for tracing recent papers that cite established seeds.

Q4. Best for foundational work?

ResearchRabbit and Litmaps, because backward citation exploration made foundational model and benchmark papers visible.

Q5. Best for themes and directions?

Elicit for structured thematic grouping; ResearchRabbit for revealing connected research directions.

Q6. Best for comparing methodologies and findings?

Elicit, because its structured extraction aligns research questions, methods, findings, and limitations.

Q7. Most unique papers?

Semantic Scholar in the reconstructed evidence set, contributing five unique records.

Q8. Most irrelevant results?

Semantic Scholar produced the most tangential results under broad queries, although filtering corrected this. Network tools instead risked drift from weak seeds.

Q9. Did four tools improve the review?

Yes. Search, network expansion, chronology, and structured synthesis are complementary; relying on one tool would reduce either coverage or analytical structure.

Q10. Future combination?

Use Semantic Scholar first, ResearchRabbit and Litmaps second for expansion and lineage, Elicit third for evidence extraction, then verify primary records manually before writing.

Part H — Comparison of Review-Output Support

Dimension	ResearchRabbit	Litmaps	Semantic Scholar	Elicit
Organisation of review	3	4	3	5
Coverage of major themes	4	5	5	4
Depth of synthesis	3	3	3	5
Identification of gaps	3	4	4	5
Usefulness for academic writing	3	4	4	5
Overall quality	4	4	4	5

The tools should not be treated as autonomous authors. Their strongest role is upstream evidence organization. Final review quality depends on manual inclusion decisions, source reading, claim-level verification, and thematic synthesis.

Part I — Prompt and Query Record

ID	Tool	Exact prompt / search query	Purpose	Used?
PR01	ResearchRabbit	Seed papers: BLOOM (2022), XTREME (2020), mT5 (2021), Cross-lingual consistency of factual knowledge (2023). Explore similar, earlier, and later work related to multilingual LLM reliability and scientific	Expand citation/author network	Yes

ID	Tool	Exact prompt / search query	Purpose	Used?
		text analysis.		
PL01	Litmaps	multilingual large language models AND factuality AND hallucination AND scientific literature analysis	Map chronological and citation development	Yes
PS01	Semantic Scholar	multilingual large language model factuality cross-lingual consistency low-resource languages	Find directly relevant influential papers	Yes
PS02	Semantic Scholar	large language models automated literature review citation hallucination scientific text analysis	Find recent review-automation evaluations	Yes
PE01	Elicit	How does the reliability of large language models vary across languages when they summarize, extract, or synthesize scientific literature?	Compare methods, findings, and limitations	Yes
PE02	Elicit	Which retrieval, prompting, fine-tuning, calibration, and human-verification methods reduce multilingual hallucination in scientific analysis?	Identify mitigation evidence and gaps	Yes

Part J — Student's Critical Evaluation

The four tools are most useful when assigned distinct roles rather than judged as interchangeable search engines. In this topic, Semantic Scholar is the strongest overall entry point because it supports precise topical searching and makes influential and recent papers comparatively easy to locate. Queries combining multilingual LLMs, factuality, hallucination, and scientific literature analysis lead directly to benchmark papers and application evaluations. Its weakness is breadth without context: broad searches can mix general hallucination studies, multilingual NLP papers, and domain-specific scientific applications without explaining how the literature developed.

ResearchRabbit adds that missing relational context. Starting from trusted seed papers such as XTREME, mT5, BLOOM, and the EMNLP study on cross-lingual factual consistency makes it easier to expand toward earlier multilingual representation research and later benchmark work. It is particularly valuable for identifying clusters and recurring authors. However, its output is strongly seed-dependent. A narrow or weak seed set can trap the search in one citation community and miss relevant work that uses different terminology.

Litmaps is most valuable for chronology and citation lineage. It helps distinguish foundational multilingual-model papers from later factuality benchmarks and application studies. For a rapidly changing topic, forward-citation tracing is useful because recent benchmark papers may not rank highly in conventional searches. Its limitations are incomplete coverage for very recent preprints and the possibility that citation proximity is mistaken for topical relevance.

Elicit contributes most during evidence extraction and synthesis. Organizing studies by question, method, dataset, result, and limitation reveals methodological patterns that are difficult to see in an unstructured paper list. It also exposes gaps—for example, many studies evaluate general multilingual reasoning or English-language literature automation, but few evaluate claim verification and citation grounding across scientific domains and languages simultaneously. Nevertheless, extracted summaries cannot replace reading. Technical details, qualifiers, and negative results require checking in the original paper.

Overlap among the tools is substantial for influential papers such as XTREME, mT5, BLOOM, retrieval-augmented generation, and major factuality studies. This repetition is useful as a relevance signal but can create a false impression of coverage if all tools index the same highly cited English-language literature. The greatest failure across the workflow is therefore not a lack of papers, but underrepresentation of non-English scientific corpora and evaluations.

Human judgement remains essential at every stage. Researchers must formulate queries, choose seeds, remove duplicates, decide relevance, inspect persistent identifiers, distinguish metadata authenticity from claim support, and organize the final review thematically. The supplied Lab 1 audit demonstrates why this matters: professional-looking citations can contain metadata errors, and genuine papers can still be used to support inaccurate claims.

For future work, I recommend a staged workflow: begin with Semantic Scholar for a controlled core set; use ResearchRabbit and Litmaps to expand backward, forward, and sideways; use Elicit to build an evidence matrix; then return to publisher, DOI, arXiv, PubMed, or ACL records for verification. Draft only after the evidence matrix is stable, and verify every quantitative or comparative claim against the source text. This workflow uses each tool for its comparative advantage while keeping scholarly judgement and source evidence central.

Part K — Recommended Workflow

Stage	Recommended tool(s)	Reason
Topic exploration	Semantic Scholar + Elicit	Refine concepts, synonyms, and research question.
Initial literature discovery	Semantic Scholar	Build a precise, high-relevance seed set.
Foundational/seminal literature	ResearchRabbit + Litmaps	Trace backward citations and influential clusters.
Recent literature	Semantic Scholar + Litmaps	Combine recency filtering with forward citations.
Related literature expansion	ResearchRabbit	Explore author, citation, and similarity networks.
Literature comparison/synthesis	Elicit + manual evidence matrix	Align methods, findings, datasets, and limitations.
Research-gap identification	Elicit + Litmaps	Contrast extracted limitations with chronological coverage.
Final review writing	Manual synthesis with source verification	Write thematically; validate identifiers and exact claim support.

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Final Checklist and Declaration

- ☑️ Assigned topic used consistently.
- ☑️ Four-tool workflow and prompts documented.
- ☑️ Final evidence set contains 20 scholarly papers.
- ☑️ Review is thematic and analytical.
- ☑️ Comparative ratings, observations, critical evaluation, and recommended workflow included.
- ☑️ Citation-audit lessons incorporated without repeating the entire Lab 1 procedure.

Declaration status: Not signed in this generated draft. Sign only after confirming that the tool-use descriptions, counts, prompts, and ratings match your actual ResearchRabbit, Litmaps, Semantic Scholar, and Elicit sessions.

Field	Response
Student name	Prakhar Srivastava
Roll number	PNS2026002
Topic ID	T6
Date	11 September 2026
Signature	_____